

Property	Image $\rightarrow$ text carrier	Text $\rightarrow$ PNG carrier
Canonical payload	16×16 grayscale (L8), row-major; 256 raw pixel bytes	Literal strict UTF-8; allowed 32–128 B (held out: 58–128 B)
Delivered carrier	UTF-8 file; at most 2,048 carrier tokens	Lossless PNG, 32×31 RGB8; 992 pixels / 2,976 channel values
Model / checkpoint	Llama 3 8B Instruct LLM; QuantFactory Q4_K_M GGUF; 86c8ea6c8b75	CIFAR-10 pretrained PixelCNN++ autoregressive image model; 5 residual blocks, 160 filters, 10 logistic mixtures; a5ed558f6d40
Inference profile	llama-cpp-python 0.3.23; full 33/33 eligible layers on CUDA; float64 probability bookkeeping	PyTorch 2.5.1+cu124; float32 CUDA forward; float64 discrete RGB conditional/mixture posterior probabilities
Shared context	Frozen prompt1: temperate-forest field journal; prompt tokenization separate from carrier	Frozen row1: 1×32 RGB8 (96 B), prepended privately to a 32×32 model canvas
Packet and framing	292 B = 12 B nonce + encrypted (8 B header + 256 B slot) + 16 B tag; AES-256-GCM; zero slot padding	Same 292 B / 2,336-bit packet; one sealed packet shared across methods in each payload/context group
Coding	Fixed radix 16; entropy-gated radix 16; finite-precision arithmetic A1 (32 bits)	Same three methods; RGB channel alphabet; raster order, then R/G/B
Eligibility / gate	Top 256 before complete-prefix filtering; strict $H > 0.598397$ bits	Observable values 0–255; strict $H > 3.215951$ bits
Stopping / completion	Known 2,336-bit target; exactly 32 ordinary tail tokens after packet completion, within cap	Known packet target; ordinary completion through all 2,976 delivered channels
Exact recovery	Fresh receiver reconstructs tokens from saved UTF-8; authenticated parsing plus independent equality of all canonical pixels	Fresh receiver reconstructs symbols from saved PNG pixels; authenticated parsing plus independent literal source-byte equality
Distinct payloads	Development 20×2 prompts; held out $20 \times$ prompt1	Development 20×2 rows; held out $20 \times$ row1
V2 allocation	60 stego units; 20 independent control traces with matched prefixes shared across methods	60 stego units; 20 independent full-PNG controls shared across methods

Experimental specification. Frozen scientific profiles, not the later optional GPU graph execution mode. Development uses 20 distinct payloads per direction; fixed-arm cross-context tests do not imply that every development payload had every method. V2 has 20 groups per direction, three paired methods and one context. Checkpoint hashes are abbreviated to 12 hexadecimal characters; gate values are rounded here only. Exact float64 thresholds, checkpoint hashes, contexts, runtime identities and source provenance are in data/protocol_identities.json. The 36 framing bytes and slot padding are different overheads. Text and image neural models are different autoregressive backends. Exactness concerns canonical source bytes, not pre-resize images or arbitrary source containers.